

Enterprise Windows Active Directory Infrastructure Lab (Version 1)

Project Overview

This project demonstrates the deployment of an enterprise Windows infrastructure using Windows Server 2022 and a Windows 11 client in Oracle VirtualBox. The environment includes Active Directory Domain Services (AD DS), DNS, DHCP, and domain-based authentication.

Objectives

- Deploy Windows Server 2022 as a Domain Controller.
- Configure AD DS.
- Configure DNS and DHCP.
- Join a Windows 11 client to the domain.
- Validate enterprise networking.

Technology Stack

Windows Server 2022
Windows 11
Oracle VirtualBox
AD DS
DNS
DHCP
TCP/IP (IPv4)

Architecture

A Windows Server 2022 Domain Controller (DC01) hosts Active Directory, DNS, and DHCP services. A Windows 11 client (CLIENT01) joins the domain and receives network settings from DHCP.

Implementation Workflow

1. Install VirtualBox
2. Create Windows Server VM
3. Install AD DS
4. Promote to Domain Controller
5. Configure DNS
6. Configure DHCP
7. Create Windows 11 Client VM
8. Join CLIENT01 to the domain
9. Validate infrastructure.

Validation

AD Users and Computers
DNS Manager
DHCP Address Leases
Client Domain Joined
ipconfig /all
ping DC01

Group Policy Management

Skills Demonstrated

Windows Server Administration

Active Directory Administration

DNS Administration

DHCP Administration

Enterprise Networking

Troubleshooting

Future Enhancements

OPNsense Firewall

Wazuh SIEM

Splunk

Sysmon

Windows Event Forwarding